

Theorem 2.29 The Pumping Lemma for Regular Languages

Suppose L is a language over the alphabet Σ . If L is accepted by a finite automaton $M = (Q, \Sigma, q_0, A, \delta)$, and if n is the number of states of M , then for every $x \in L$ satisfying $|x| \geq n$, there are three strings u , v , and w such that $x = uvw$ and the following three conditions are true:

1. $|uv| \leq n$.
2. $|v| > 0$ (i.e., $v \neq \Lambda$).
3. For every $i \geq 0$, the string $uv^i w$ also belongs to L .

- The most common application of this lemma is to show that a language *cannot* be accepted by an FA, by **showing that** it doesn't have the property described in the pumping lemma.
- This can be done by showing that **there exists** some $x \in L$ satisfying $|x| \geq n$, such that **every choice** for the three strings u , v , and w such that $x = uvw$ **fails** to hold that the following three conditions are true:

1. $|uv| \leq n$.
2. $|v| > 0$ (i.e., $v \neq \Lambda$).
3. For **every** $i \geq 0$, the string $uv^i w$ also belongs to L .

Some examples of application of the lemma to show that a language cannot be accepted by an FA
(hence, not a Regular Language)

Let L be the language $AnBn$ defined as follows:

$$L = \{a^i b^i \mid i \geq 0\}$$

It would be surprising if $AnBn$ could be accepted by an FA. If the beginning input symbols are a 's, a computer accepting L surely needs to remember how many of them there are, because otherwise, once the input switches to b 's, it won't be able to compare the two numbers.

Let's prove that $AnBn$ is not a regular language using the Pumping Lemma.

Suppose for the sake of contradiction that there is an FA M having n states and accepting L . **We** choose $x = a^n b^n$. Then $x \in L$ and $|x| \geq n$. Therefore, by the

pumping lemma, there should be strings u, v , and w such that $x = uvw$ and the conditions 1–3 in the theorem are true.

Because $|uv| \leq n$ (by condition 1) and the first n symbols of x are a 's (because of the way **we** chose x), all the symbols in u and v must be a 's. Therefore, $v = a^k$ for some $k > 0$ (by condition 2).

We can get a contradiction from statement 3 by using **any** number i other than 1, because $uv^i w$ will still have

- exactly n b 's, but
- will no longer have exactly n a 's.

The string $uv^2 w$, for example, is $a^{n+k} b^n$, obtained by inserting k additional a 's into the first part of x .

This is a contradiction, because

- the pumping lemma says $uv^2 w \in L$, but
- $n + k \neq n$, therefore, $uv^2 w \notin L$

We have shown that there exists an $x \in L$ satisfying $|x| \geq n$, that does not have the property described in the pumping lemma, hence, L is not a regular language.

Let L be the language defined as follows:

$$L = \{a^{i^2} \mid i \geq 0\}$$

i.e., the strings in L comprise of a 's only, and the length of every string is a perfect square.

Let's prove that L is not a regular language using the Pumping Lemma.

Suppose for the sake of contradiction that there is an FA M having n states and accepting L .

We choose $x = a^{n^2}$.

Then $x \in L$ and $|x| \geq n$. Therefore, by the pumping lemma, there should be strings u, v , and w such that $x = uvw$ and the conditions 1–3 in the theorem are true.

Because x comprises of a 's only, $v = a^k$ for some $k > 0$ (by condition 2). Moreover, as $|uv| \leq n$ (by condition 1) we have that $k \leq n$.

We can get a contradiction from statement 3 by using $i = 2$, as follows:

$$n^2 = |uvw| < |uv^2w| = n^2 + k \leq n^2 + n < n^2 + 2n + 1 = (n + 1)^2$$

As, length of uv^2w is strictly greater than n^2 but strictly smaller than $(n + 1)^2$, it cannot be a perfect square.

This is a contradiction, because

- the pumping lemma says $uv^2w \in L$, but
- $n^2 + k$, is not a perfect square, therefore, $uv^2w \notin L$

We have shown that there exists an $x \in L$ satisfying $|x| \geq n$, that does not have the property described in the pumping lemma, hence, L is not a regular language.